RW–9981 **521704**

M.Sc. DEGREE EXAMINATION, APRIL 2015

Physics

CONDENSED MATTER PHYSICS

(CBCS – 2006 onwards)

Time : 3 Hours Maximum : 100 Marks

Answer **all** the questions. (5 × 20 = 100)

1. (a) (i) Explain in detail the two and three dimensional lattices with suitable diagrams.

(ii) Discuss the Hexagonal closed packed structure.

Or

(b) Discuss in detail the different types of chemical bonds in crystals.

2. (a) (i) What are phonons?

(ii) Derive the dispersion relation for two atoms permittive cell.

Or

(b) (i) What is Hall effect?

(ii) Explain the energy band on the basis of Kronig Penney model.

3. (a) (i) Define polarizability.

(ii) Derive an expression between the microscopic and macroscopic property of a dielectric material.

Or

(b) (i) Classify the ferroelectric crystals.

(ii) Discuss the Landau theory of phase transition.

4. (a) (i) How are dia, para and ferro magnetic materials classified?

(ii) Explain the quantum theory of paramagnetism.

Or

(b) (i) What are magnons?

(ii) Derive an expression for dispersion relation for antiferro agnetic material.

5. (a) (i) Distinguish between type I and type II superconductors.

(ii) Explain the BCS theory of superconductivity.

Or

(b) (i) What is meant by tunneling in superconductors?

(ii) Explain AC Josephson effect for superconducting material.

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RW–9967 **521201**

M.Sc. DEGREE EXAMINATION,
APRIL 2015

Second Semester

Physics

QUANTUM MECHANICS – I

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What is meant by wave-particle duality?

2. What is tunnel effect?

3. What are the allowed energy eigen values of a rigid rotator?

4. State any two features of energy levels of hydrogen atom.

5. Define perturbation and unperturbed part of a Hamiltonian.

6. State the three steps involved in the WKB method.

7. What do the Einstein's coefficients describe?

8. Define spontaneous and stimulated emissions.

9. If the scattering amplitude is $(\cos\theta)/5$, determine the total scattering cross – section.

10. What are partial waves?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Find the potential corresponding to

$$\Psi = \left(\frac{1}{\pi}\right)^{1/4} e^{-x2/2} \text{ and } E = \hbar w/2.$$

Or

(b) State the postulates of quantum mechanics.

12. (a) The eigen function of 1s electron is $\left(1/\left(\pi a_0^3\right)\right)^{\frac{1}{2}} e^{-r}/a_0$. Calculate the probability of finding the electron in a sphere of radius $r = a_0$.

Or

(b) Set up the schrödinges equation for a rigid rotator.

13. (a) Determine the first–order correction to energy eigen value for nondegenerate energy levels.

Or

(b) Find the ground state energy of deuteron by the method of variation.

14. (a) Establish the connection between the Einstein's coefficients.

Or

(b) Obtain an expression for the probability for induced emission.

15. (a) Applying Born approximation, find the scattering amplitude for a screamed coulomb potential.

Or

(b) Explain the optical theorem.

Part C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. State and prove Ehrenfest's theorem.

17. Solve the schrödinger equation to determine the energy eigenvalues and eigenfunctions of three-dimensional harmonic oscillator.

18. Explain Stark effect applying a suitable perturbation theory.

19. Obtain the transition rate per unit time for a system subjected to a constant perturbation during the time interval 0 to T.

20. Outline the method of partial wave analysis.

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RW–9968 **521202**

M.Sc. DEGREE EXAMINATION, APRIL 2015

Second Semester

Physics

MATHEMATICAL PHYSICS – II

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Write the Laurent series of $\frac{1}{z(z+1)}$ valid bear $z=0$.

2. Identify the nature of singularities of $\frac{\sin z}{z}$ and $\sin\left(\frac{1}{z-c}\right)$.

3. Give the physical interpretation of the solution $u(x,t)=\frac{1}{2}[f(x-ct)+f(x+ct)]$ of the wave equation.

4. Define Green's function.

5. State the meaning of completeness of eigen functions.

6. Write the equation $x^2y''+xy'+(x^2-9)y=0$ in the Strum – Liouville form.

7. Find the contraction of (A^iB_j).

8. What are Cartesian and non Cartesian tensors?

9. What are the Abelian groups?

10. Define the character of a representation.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) State and prove Cauchy – Rieman conditions.

Or

(b) Prove Cauchy's integral formula.

12. (a) Write the Laplace and Poisson equations and explain the difference between them.

Or

(b) Determine the Green's function for the boundary value problem $y''+y=-f(x)$ with $y(0)=y(L)=0$.

13. (a) Outline the method of construction of an orthogonal set of eigen functions of a Hermitrain operator.

Or

(b) Establish that the Sturm – Liouville operator is Hermitrain.

14. (a) Enumerate the covariant formulation of electrodynamics.

Or

(b) Express velocity and acceleration in four – vectors notation.

15. (a) Construct the characteristic table of C_{3v}.

Or

(b) Write short note on crystallographic point groups.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Evaluate $\int_0^\infty \frac{(\log x)^2}{(1+x^2)}\, dx$.

17. Construct the solution of Laplace equation.

18. Determine the eigenvalues and eigen functions of the Sturm – Liouville equation $y''+\lambda y=0$ for $\lambda=0$, <0 and >0 subject to the conditions $y(0)=y(L)y'(0)=y'(L)$.

19. State and prove quotient law and then find the rank of A such that $A(i,j)\, B^j = C^i$.

20. Prove the orthogonality relation for irreducible representation.

RW–9969 **521203**

M.Sc. DEGREE EXAMINATION, APRIL 2015

Second Semester

Physics

ELECTROMAGNETIC THEORY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Using the divergence theorem obtain the Gauss's law.

2. Find the magnitude of the magnetic induction at a point R from the long straight-wire carrying a current I.

3. What are the significances of Brewster's angle?

4. What are the kinematic and dynamic properties of the reflection of light at a plane surface between two different media?

5. Distinguish anomalous dispersion from normal dispersion.

6. Define the scattering parameters.

7. What are called modes?

8. Mention the applications of resonators.

9. What does magneto hydrodynamics refer?

10. What are plasma waves?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Obtain the solution of electrostatic boundary-value problem using Green's theorem.

Or

(b) Show that energy flow associated with an electromagnetic field is proportional to curl of $\vec{E}$ and $\vec{H}$.

12. (a) A plane electromagnetic wave in free space is incident upon the plane boundary of a thick dielectric at an angle of 56° to the normal to the boundary. The wave is polarized in the plane of incident. What is the value of the relative permittivity of the dielectric if there is no reflected wave?

Or

(b) Discuss the phenomenon of total internal reflection.

13. (a) Derive Classius-Mossotti relation.

Or

(b) Explain the phenomenon of polarization of scattered light.

14. (a) Explain the working principle of Gunn diode.

Or

(b) Find the modes of 3 cm wavelength radar waves that would be propagated in a wave guide of rectangular cross-section with $a = 1$ cm and $b = 2$ cm.

15. (a) Explain the motion of charged particles in crossed electric and magnetic fields for the case of drift velocity equal to $(\vec{E} \times \vec{B}) / B^2$.

Or

(b) Explain the conditions for the existence of plasma.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Discuss the problem of propagation of electromagnetic waves in a non-conducting isotropic medium.

17. Obtain the relation amplitudes of reflected and refracted electric field for the case of electromagnetic field parallel to the plane of incidence.

18. Discuss the propagation of electromagnetic waves in a dilute gas.

19. Explain the occurrence of TE and TM modes in a resonant cavity.

20. Obtain the equations of hydrodynamics that a plasma must obey.

RW–9970 **521506**

M.Sc. DEGREE EXAMINATION, APRIL 2015

Second Semester

Physics

Elective — DIGITAL ELECTRONICS PRINCIPLES

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Give an example for the Double Dabble method of converting decimal to binary.
2. What is an octal odometer?
3. Which gates are called universal gates?
4. What is meant by 'don't care condition' in Karnaugh mapping?
5. What is an encoder?
6. What is the function of binary divider?
7. Define the modulus of a counter.
8. How is a shift register used as a ROM?
9. Give the truth table of S-R and J-K flip flops.
10. What is meant by race around condition in J-K flip flops.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

All questions carry equal marks.

11. (a) Convert the following two octal numbers to their decimal equivalents.

(i) 216 (ii) 65

Or

(b) Give an account of alphanumeric codes with suitable examples.

12. (a) Discuss minimization in SOP form using Karnaugh map with suitable examples.

Or

(b) State and prove De Morgan's theorems.

13. (a) What is a comparator? Explain its working with an example.

Or

(b) Explain the working of a binary multiplier with a diagram.

14. (a) Give the logic circuit and timing diagram of an edge triggered J-K flip flop.

Or

(b) Explain the working of clocked R-S flip flop with a diagram.

15. (a) Show the circuit of a down counter and explain how it counts from 1111 to 0000.

Or

(b) Draw the circuit diagram of a three bit binary up-down counter and mention the conditions for counting up and down counting.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

All questions carry equal marks.

16. What are BCD numbers? What are its advantages? Explain the method of addition of the following digital numbers using BCD addition.

17. Reduce the function $F = \Sigma m\ (2,3,5,7,9,11,12,13,14,15)$ by Karnauch mapping, and implement in AND-OR-Invert logic.

18. Distinguish between half adder and a full adder circuit with reference to their capabilities to perform the binary addition.

19. Explain the operation of a 4-bit shift register. How is this system used as divide 4 and divide 8 counters?

20. What are synchronous and asynchronous counters? Explain the working of a decade counter.

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RW–9971 **521401**

M.Sc. DEGREE EXAMINATION, APRIL 2015

Fourth Semester

Physics

CONDENSED MATTER PHYSICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Construct Wigner-Seitz cell for an oblique lattice in two dimensions.

2. What are inert crystals?

3. What are the normal modes of vibrations?

4. What is Bloch wave?

5. What is meant by polarisation catastrophe?

6. Why are dielectric materials are insulators at ordinary temperatures?

7. What is the reason for the negative susceptibility in a diamagnetic material?

8. What are Magnons?

9. Explain the infrared properties of a typical superconducting material.

10. Define the Cooper pairs.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

All questions carry equal marks.

11. (a) How many Bravais lattices are there in three dimensions? Draw diagrams for each.

Or

(b) Is five fold symmetry possible in crystals? Mention recent findings about five fold symmetry.

12. (a) Explain the elementary theory of semiconductors with necessary diagrams.

Or

(b) Obtain the dispersion relation for elastic waves in a linear monoatomic chain taking into account the nearest neighbour interaction.

13. (a) Explain the frequency dependence of the several contributions of Polarizability.

Or

(b) Discuss the temperature dependence of dielectric constant of materials. List the applications of dielectric materials.

14. (a) What are soft magnetic materials? Discuss its characteristics and its use.

Or

(b) Outline the Langevin's theory of paramagnetism and obtain an expression for Susceptibility.

15. (a) Show that London equation leads to Meissner effect and discuss the importance of London penetration depth.

Or

(b) In insulators, the energy gap is tied to the lattice whereas in a superconductor, it is tied to the Fermi gas of electrons. Justify this statement with the necessary theory.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

All questions carry equal marks.

16. What are the different types of bonding exists in single crystals? List them. Explain in detail the nature, properties and uses of any three with examples.

17. Explain Hall effect. Obtain suitable relations for the Hall coefficient, Hall angle. Describe the experimental arrangement for the measurement of Hall voltage.

18. Explain pyrolectric effect and discuss its mechanism with suitable examples. Discuss the properties of pyroelectric crystals.

19. What are ferromagnetic domains? How can one observe the domain structure? Obtain an expression for the total Bloch wall energy per unit area.

20. Discuss the principle of DC Josephson effect. Prove that with no applied voltage a DC current flows across the junction.

RW–9972 **521402**

M.Sc. DEGREE EXAMINATION, APRIL 2015

Fourth Semester

Physics

NUCLEAR AND PARTICLE PHYSICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. List the properties of beta particles.

2. What is called internal conversion?

3. Write down the salient features of the Collective model.

4. What are magic numbers?

5. What are Majorana forces?

6. Why dineutron and diproton are not found in a stable stationary state?

7. What is meant by resonance scattering?

8. Mention the source of stellar energy.

9. List the pions and their antiparticles.

10. What is isospin?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

All questions carry equal marks.

11. (a) Explain in detail the parity violation during beta decay process.

Or

(b) Discuss the multipolarity of gamma radiations with the diagrams.

12. (a) Discuss the evidence that led to the shell model. What are its predictions?

Or

(b) Derive and explain an expression for nuclear stability.

13. (a) Explain in detail the four types of basic interactions and their significance.

Or

(b) Distinguish between nuclear fission and fusion with examples.

14. (a) What is level width of resonance? How is it associated with the life time of the state?

Or

(b) Obtain an expression for the critical size of a reactor.

15. (a) State and explain C.P.T theorem.

Or

(b) One K meson decays into two spin zero neutral pions. Obtain the possible values of the K meson spin.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

All questions carry equal marks.

16. Give Fermi's theory of beta decay process and show that it explains the momentum spectrum of beta particles.

17. Derive Weizacker's semi empirical mass formula and explain the each term in it. Write down its merits and demerits.

18. Describe the basic ideas in Yukawa's meson theory of nuclear force. Assuming the range of the nuclear force is of the order of nuclear dimensions, calculate the mass of the exchanged particle by using Heisenberg's uncertainty principle.

19. Explain the process of thermo nuclear reactions with suitable examples. How controlled thermo nuclear reactions can be achieved? Explain.

20. What is a Baryon? Explain the formation of Baryon octet with a diagram. What is its significance?

RW–9973 **521403**

M.Sc. DEGREE EXAMINATION, APRIL 2015

Fourth Semester

Physics

MATERIALS SCIENCE

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

All questions carry equal marks.

1. Distinguish elastic behavior with viscoelastic behavior.

2. Define polymer and list out its classification.

3. What the principle of Penning gauge?

4. Define epitaxial growth.

5. Write the working principle of He-Ne laser.

6. What is Kerr effect?

7. What are metal matrix composites?

8. What are the different processing methods of ceramics?

9. Define Pseudo elasticity.

10. What are shape memory alloys?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain in detail the different structures of polymers with example.

Or

(b) Discuss in detail the corrosion behavior in metals.

12. (a) Describe the deposition of thin films by thermal evaporation.

Or

(b) Describe in detail the quartz —crystal balance to monitor the thickness of thin films.

13. (a) Explain in detail the second harmonic generators and electro-optic materials.

Or

(b) Give a detailed account on photorefractive materials and a few important applications.

14. (a) Discuss in detail the ceramic-matrix composites and their applications.

Or

(b) Give the electronic and thermal properties of matrix composites.

15. (a) Describe in detail any one method to prepare amorphous and glassy materials.

Or

(b) Explain in detail the fabrication of piezoelectric MEMS.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain in detail the structure and properties of different types of polymers and their applications.

17. Discuss any two deposition techniques used for the fabrication of thin films.

18. Explain the principle, working and applications of Nd- YAG lasers in detail. Also explain the applications of lasers in materials research.

19. Give a detailed account on carbon matrix composites and their applications.

20. Describe in detail the fabrication of MEMS using silicon, porous silicon and silicon oxide.

RW–9974 **521510**

M.Sc. DEGREE EXAMINATION, APRIL 2015

First Semester

Physics

CRYSTAL GROWTH

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define nucleation.

2. Write the type of crystal symmetry.

3. Explain Bragg's law.

4. Write the role of supersaturation in crystal growth.

5. Define the working principle of sol-gel process.

6. State syneresis process.

7. Write the effect of convection in melt.

8. Describe the phases of matter.

9. Depictchoice of flux.

10. List the types of crystal defects.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain the importance of crystal growth.

Or

(b) Write a detailed note on grain boundary and dislocation density.

12. (a) Describe Meir's solubility diagram.

Or

(b) Discuss in detail the U tube and straight tube methods.

13. (a) Write a detailed note on zone refining technique.

Or

(b) Explain in detail the different flux growth techniques.

14. (a) Explain the principle and applications of chemical vapor deposition.

Or

(b) Discuss in detail the crystal systems with necessary diagram.

15. (a) Describe the optical and mechanical properties of crystals.

Or

(b) Discuss the importance of crystal cutting and polishing.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe in detail the crystal growth by low temperature solution technique .

17. Explain the principle and properties of gel growth technique and also write their advantages and disadvantages over other methods.

18. Discuss in detail the melt growth techniques and their limitation.

19. How to determine the orientation of crystal by using X-Ray method?

20. Discuss in detail the crystal defects.

RW–9975 **521101**

M.Sc. (Arrear) DEGREE EXAMINATION, APRIL 2015

First Semester

Physics

CLASSICAL MECHANICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. State the principle of virtual work.

2. Define cyclic coordinates.

3. Mention the principle of least action.

4. What is phase space?

5. Give the relation between angular momentum and kinetic energy.

6. What are Eulerian angles?

7. State the postulates of special theory of relativity.

8. Give the expression for mass of a moving particle.

9. What is meant by propulsive efficiency?

10. Give the reason for staging.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Obtain the Lagrangian equation of motion of a simple pendulum.

Or

(b) State and explain D'Alembert's Principle.

12. (a) Solve the simple Harmonic oscillator problem using Hamilton Jacobi theory.

Or

(b) Explain the significance of Hamilton's characteristic function.

13. (a) Obtain the angular velocities in terms of Euler angles.

Or

(b) Derive the expression for the period of compound pendulum.

14. (a) Derive the relativistic expression for kinetic energy.

Or

(b) Given an account of evidences in mass energy relation.

15. (a) Explain the optimization of a multistage rocket.

Or

(b) Briefly explain the performance of a single stage rocket.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Derive Lagrangian equation of motion for a conservative system using D'Alembert's principle.

17. State and prove Liouville's theorem.

18. State and prove the Euler's theorem for the motion of a rigid body.

19. Obtain the Lorentz transformation equations and discuss its consequences.

20. Derive the equation of motion of variable mass of particle.

RW–9976 **521102**

M.Sc. (Arrear) DEGREE EXAMINATION, APRIL 2015

Physics

First Semester

MATHEMATICAL PHYSICS – I

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Show that a $x(bxc)+bx(cxa)+cx(axb)=0$.

2. Calculate the differentiation of scalar triple product.

3. List out the properties of conjugate transposed matrices.

4. If A and B are Hermitian, show that $AB + BA$ is Hermitian and $AB - BA$ is Skew-Hermitian.

5. Calculate $\int_0^{\frac{\pi}{2}} \sqrt{\tan\theta}\, d\theta$ using gamma function.

6. Prove that $\cos x = J_0(x) - 2J_2(x) + 2J_4(x).....$

7. Write down the Dirichlet's conditions.

8. Find the inverse Laplace transform of $\left[S/(S^2+a^2)^2\right]$.

9. Calculate the Binomial distribution probability for n heads in a toss of N coins $P(n,N)=\frac{1}{2^N}\frac{N!}{n!(N-n)!}$.

10. Define standard deviation.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Find $\Delta\varphi$ and $|\Delta\varphi|$ for the function $\varphi = 2xz^4 - x^2y$ at the point (2, –2, –1).

Or

(b) State and prove stokes theorem in space.

12. (a) Prove that any two given vectors corresponding to two distinct eigenvalues of a Hermitian matrix are orthogonal.

Or

(b) Write a short note on unitary matrix and orthogonal matrix.

13. (a) Prove that $L_n(2x) = n!\sum_0^\infty \frac{2^{n-m}(-1)^m}{m!(n-m)!} L_{n-1}(x), \quad n > m$.

Or

(b) Obtain the orthogonality of Legendre polynomial.

14. (a) Find the Fourier transform of $F(x) = \begin{cases} 1, & |x| < a \\ 0, & |x| > a \end{cases}$.

Or

(b) Find the solution of $\frac{\partial^2 V}{\partial x^2} - \frac{\partial^2 V}{\partial t^2} = xt$ satisfying the conditions $V = \frac{\partial V}{\partial t} = 0$ where t=0 using inverse Laplace transform.

15. (a) Show the results of throwing 12 dice, 4096 times, a throw of 4, 5 or 6 being called a success. Find the expected frequencies and compare the actual mean with those of expected distribution. Calculate the standard deviation by using, binomial distribution.

Or

(b) Explain the Poisson distribution.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Show that $[a \times b, b \times c, c \times a] = [abc] = \begin{bmatrix} a.a & a.b & a.c \\ b.a & b.b & b.c \\ c.a & c.b & c.c \end{bmatrix}$ by means of determinant.

17. Determine the eigenvalues and eigenvectors of the matrix $\begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$.

18. Prove that $\int_{-\infty}^{\infty} e^{-x^2} H_m(x) H_n(x) dx = 2^n n! \sqrt{\pi} \delta_{m,n}$ where

$$\begin{cases} \delta_{m,n} & = 0 \ when \ m \neq n \\ & = 1 \ when \ m = n \end{cases}.$$

19. If $\begin{cases} f(x) = x & for \ 0 \leq x \leq \frac{\pi}{2} \\ f(x) = \pi - x & for \ \frac{\pi}{2} \leq x \leq \pi \end{cases}$, express this function by a Fourier sine series and also by a Fourier cosine series.

20. State and prove the properties of random sampling distribution.

RW–9977 **521501**

M.Sc. (Arrear) DEGREE EXAMINATION, APRIL 2015

First Semester

Physics

ELEMENTARY NUMERICAL ANALYSIS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is known as truncation error?
2. Write the algorithm of bisection method.
3. What is the order of error is linear interpolation? Justify.
4. List the properties of divided differences.
5. How can the error is trapezoidal rule the minimized?
6. List the drawbacks of Euler's method of solving differential equations.
7. Write the condition for the applicability of Taylor's services method of solving differential equations.
8. Distinguish between direct methods and iterative methods of solving linear algebraic equations.
9. What are called the compact variants of Gaussian elimination method?
10. What is the procedure to be adapted to make the Gauss–seidel method to converge quickly?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) If $u = 4x^2y^3/z^4$ and errors in x, y, z by 0.001, compute the relative maximum error in u when $x = y = z = 1$.

Or

(b) Derive the equations of least squares method of fitting a given data set with a straight line.

12. (a) Derive a formula for interpolation with a cubic polynomial.

Or

(b) Show that the nth divided differences of a polynomial of n-th degree are constant.

13. (a) Derive the numerical formula for computing the integration of a function using the trapezoidal rule.

Or

(b) Explain the procedure of Richardson extrapolation and hence deduce the corresponding computational formula.

14. (a) Apply the second order Runge – Kutta method to find an approximate value of $y(x)$ when $x = 0.2$, given that $(dy/dx) = x + y$ and $y = 1$ when $x = 0$.

Or

(b) Explain the numerical procedure of computing the inverse of a matrix.

15. (a) Explain the scheme of LU factorization to solve a system of linear equations.

Or

(b) Deduce the Gauss – Seidel iterative formulas to solve a system of linear equations.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Find the root of the equation $xe^x = \cos x$ using the Newton method and secant method, correct to four decimal places, and compare the results.

17. Determine $f(x)$ as a polynomial in x for the following data:

x:	-4	-1	0	2	5
$f(x)$:	1245	33	5	9	1335

18. Obtain solution of the equation $(dy/dx) = x + 1\sqrt{y}1$, with initial conditions $y = 1$ at $x = 0$, for the range $0 \le x \le 0.6$ in steps of 0.2, using Euler's method and Euler's modified method and compare the results.

19. Apply Gauss elimination method to solve the equations
$x + 4y - z = -5,$
$x + y - 6z = -12,$.
and $3x - y - z = 4$

20. Solve, by Jacobi's method, the following equations:

$20x + y - 2z = 17,$
$3x + 20y - z = -18,$
and $2x - 3y + 20z = 25$

RW–9978 **521302**

M.Sc. (Arrear) DEGREE EXAMINATION, APRIL 2015

Third Semester

Physics

QUANTUM MECHANICS — II

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are Pauli's spin matrices?

2. Calculate the value of $(\sigma_x + \sigma_y + \sigma_z)^2$.

3. Distinguish symmetric and antisymmetric wave function.

4. Define Hilbert space.

5. Is Thomas-Fermi method suitable for heavy atoms?

6. What is meant by jj coupling scheme?

7. Express the Dirac's Hamiltonian.

8. Show that "Pauli spin matrices are anti-commutes with each other".

9. Define second quantisation.

10. Differentiate Dirac field and vector field.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Find the commutation relations of angular momentum using $\vec{L} = \vec{r} \times \vec{P}$.

Or

(b) Write a note on angular momentum matrices.

12. (a) Briefly explain the vectors of Dirac's ket and bra notations.

Or

(b) Explain the Schrodinger picture.

13. (a) Explain the Thomas — Fermi statistical model of a heavy atom.

Or

(b) Give an account of corrections in central field approximation.

14. (a) Derive the Klein — Gordan equation.

Or

(b) Write a note on negative energy states.

15. (a) Derive the energy eigenvalue for the system of Bosons field quantization.

Or

(b) Explain the field quantization of Fermions for non-relativistic field.

Part C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. Derive the Clebsch — Gordon coefficient for j = ½ system.

17. Write a note on the following:

    (a) Heisenberg picture

    (b) Interaction picture.

18. Describe the self-consistent Hartree — Fock field method.

19. Explain the Dirac relativistic equation for free electron.

20. Describe the quantization of electromagnetic field in vacuum.

RW–9979 **521303**

M.Sc. (Arrears) DEGREE EXAMINATION, APRIL 2015.

Third Semester

Physics

MICROPROCESSOR AND ELECTRONIC INSTRUMENTATION

(CBCS 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What do you mean by "One byte" instruction.

2. Mention the use of data transfer group instructions.

3. What happens when NOP instruction is executed?

4. State the use of stack.

5. What is the function of DPTR in 8051 microcontroller?

6. How many input and output ports can be connected in direct I/O.

7. What are the operating modes of Intel 8255?

8. State the function of stepper motor.

9. List the uses of D/A converter.

10. Mention a few sensors and transducers used in physical measurements.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

All questions carry equal marks.

11. (a) Explain the functions of ALU in 8085.

Or

(b) Discuss the special nature of accumulator.

12. (a) Explain the use of DAA instructions with illustrations.

Or

(b) Describe the software development tools available in 8085.

13. (a) Explain the addressing modes of 8051 with examples.

Or

(b) Discuss jump and call instructions in 8051.

14. (a) Explain how I/O mapped I/O scheme for allocation of address with example.

Or

(b) Explain the working of DMA controller.

15. (a) Discuss the working of sample hold circuit with diagram.

Or

(b) Explain the operation of resistive strain gauge with diagram.

Part C

(3 × 10 = 30)

Answer any **three** questions.

All questions carry equal marks.

16. Discuss in detail about the flag structure in 8085.

17. Explain with programming how a simple four junction traffic signal control is simulated.

18. With a program to pick up the smallest and the largest number among 15 numbers.

19. How 8255A is used as a Programmable Peripheral Interface (PPI). Explain with a diagram.

20. Explain with the necessary diagram the structure and operation of LVDT transducer.

RW–9980 **521301**

M.Sc. (Arrear) DEGREE EXAMINATION, APRIL 2015

Third Semester

Physics

CHEMICAL PHYSICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Differentiate atomic and hybrid orbital.

2. List out the uses of hybrid orbitals?

3. What is meant by degeneracy?

4. Why all molecules are not microwave active?

5. How many modes of vibrations are for SO_2?

6. What is meant by Fermi resonance and overtone?

7. Why are very intense light sources needed for observation of nonlinear Raman effects?

8. What is two photon absorption?

9. Define the terms ELDOR and ENDOR.

10. What is Larmor frequency?

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Write a short note on the valance band description of H_2.

Or

(b) Briefly explain the sp and sp^2 hybridization.

12. (a) The rotational constant for CO is observed to be 3.84235 cm^{-1} (J = 0). Estimate the moment of inertia and bond length.

Or

(b) Discuss the intensity of spectral lines of pure rotation spectrum.

13. (a) Explain "vibrational coarse structure progression" in electronic spectra of diatomic molecules.

Or

(b) Write a short note on :

(i) Dissociation

(ii) Predissociation.

14. (a) Explain the theory of PARS.

Or

(b) Explain multiphoton spectroscopy with a neat diagram.

15. (a) Explain the double resonance in ESR.

Or

(b) Briefly explain the role of the chemical shifts in Mossbauer spectrum.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Discuss about the molecular orbital theory for hydrogen molecule.

17. Discuss the microwave spectrum of linear polyatomic molecules.

18. Explain in detail the Fortrat parabola.

19. Discuss in detail the classical theory of hyper Raman scattering.

20. Explain how the chemical shift of a compound helps to find the molecular structure.

RW–9966 **521101**

M.Sc. (Arrear) DEGREE EXAMINATION, APRIL 2015

First Semester

Physics

CLASSICAL AND STATISTICAL MECHANICS

(CBCS – 2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. What are the generalized coordinates?

2. What is D' Alembert's principle?

3. Define cyclic coordinates.

4. What is meant by the principle of least action?

5. Define the normal modes of vibrations.

6. Draw a diagram to illustrate the linear triatomic molecule.

7. Define the root mean square velocity.

8. What is an electron gas?

9. Write down the Langevin equation.

10. State Nyquist's theorem.

Part B $(5 \times 5 = 25)$

Answer **all** the questions, choosing either (a) or (b).

11. (a) Derive Lagrange's equation of motion.

Or

(b) Explain Hamiltonian's variational principle.

12. (a) Explain canonical transformations.

Or

(b) Discuss Hamilton's equations in Poisson bracket notations.

13. (a) Discuss the generalized coordinates for rigid body motion.

Or

(b) Explain the angular momentum of a rigid body.

14. (a) Explain Bose-Einstein condensation.

Or

(b) Discuss Maxwell Boltzmann statistics.

15. (a) Derive a relation between dissipation and fluctuating force.

Or

(b) Explain the correlation function and friction force.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Discuss about the conservative and non-conservative systems.

17. Write a short note on :

    (a) Action angle variable.

    (b) Kepler's problem.

18. Discuss the motion of a symmetric top under the action of gravity.

19. Explain the Fermi Dirac distribution function.

20. Derive and explain Wiener khintehine relation.